React Day 1 & Day 2 – Detailed Interview Answers with Examples

Complete WHAT, WHY, HOW explanations with examples. Ideal for freshers and interviews.

■ Day 1 – React Basics & Virtual DOM

1. What is React and why was it created?

WHAT: React is a JavaScript library for building UI using components.

WHY: DOM manipulation becomes complex and slow in large apps.

HOW: React updates UI automatically using state changes.

```
function Hello() {
  return <h1>Hello React</h1>;
}
```

Why library, not framework?

React focuses only on the View layer. Other features are optional libraries.

React vs Vanilla JS

```
document.getElementById("root").innerHTML = "<h1>Hello</h1>";
function App() {
  return <h1>Hello</h1>;
}
```

2. What is Virtual DOM?

WHAT: Lightweight JS copy of real DOM.

WHY: DOM updates are expensive.

HOW: Diff old and new Virtual DOM and update minimal nodes.

3. What is Reconciliation?

Process of comparing old and new Virtual DOM trees.

```
items.map(item => (
  <li key={item.id}>{item.name}</li>
))
```

4. UI Update after State Change

State change → Re-render → Virtual DOM diff → Real DOM update.

```
setCount(prev => prev + 1);
```

5. Output-based Question

console.log runs on every render including parent renders.

■ Day 2 – Components & Rendering

1. Functional vs Class Components

Hooks made functional components powerful and preferred.

```
function App() {
  return <h1>Hello</h1>;
}
```

2. Props vs State

Props are read-only. State is local and mutable.

```
function Child({ name }) {
  return <h2>{name}</h2>;
}
```

3. Re-render Rules

Re-render happens on state/prop/parent update.

```
const Child = React.memo(() => <p>Hello</p>);
```

4. Lifecycle with Hooks

useEffect replaces lifecycle methods.

```
useEffect(() => {
  return () => console.log("Cleanup");
}, []);
```

5. Counter Output Explanation

Initial render = 1, each click = 1 render. Same value = no render.